Face Recognition-Based Consent Platform for Ethical Media Sharing using AR devices

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Abstract

This paper presents a novel media sharing platform that enforces consent through face recognition to protect bystander privacy in live recordings. The system leverages a Flask web backend and Microsoft HoloLens 2 augmented reality headset for live video streaming, integrating MTCNN (Multi-task Cascaded Convolutional Networks) for real-time face detection and FaceNet for face recognition. Motivated by recent privacy scandals – notably a case where Harvard students misused Meta's Ray-Ban smart glasses to dox strangers in real time.[1] Our platform aims to ensure that individuals captured in live videos or photos have given explicit permission before any content is publicly shared. Key contributions include a consent management pipeline that automatically identifies faces in a video, requests approval from identified individuals, and only allows sharing of the recording once all participants have consented. We discuss the system's architecture, ethical safeguards (privacy preservation and secure handling of biometric data), and initial findings. Our results demonstrate that the platform can capture first-person video, detect and recognize faces on-the-fly, and enforce a consent workflow with minimal latency. This consent-driven approach shows promise to raise the standard of ethical media sharing, preventing privacy breaches and empowering bystanders to control their presence in others' digital content.

1 Introduction

In the era of ubiquitous smartphones, wearable cameras, and live streaming, the privacy of individuals in public spaces has become a pressing concern. Modern social media and streaming platforms enable users to broadcast videos in real time to large audiences, often capturing bystanders without their knowledge or consent[2]. This has raised ethical and safety issues, as evidenced by public backlash against products like Google Glass in the 2010s – society grew wary that anyone could be recorded anywhere without notice.

These concerns have only intensified with the advent of advanced wearable cameras (e.g. Snap Spectacles and Meta's Ray-Ban Stories), which seamlessly integrate recording devices into everyday eyewear. Ensuring live media security and consent is therefore of paramount importance. There is a growing consensus that individuals should have control over their likeness in digital content. Unfortunately, existing platforms lack robust mechanisms to guarantee this. Once a live video is captured, it can be instantly shared or saved, and those incidentally filmed have little recourse. To address this gap, we introduce a face recognition-based consent-driven media sharing platform. The idea is straightforward: any person appearing in a live stream or photo must give consent before that content is shared beyond its creator. By combining real-time face detection/recognition with a consent approval workflow, our platform attempts to bake in privacy by design – making permission an integral prerequisite for sharing media. In this paper, we detail the design and implementation of the platform, developed at North Carolina State University. We discuss how recent events have underscored the need for such a solution, outline the system's technical components (from the HoloLens 2 capture device to the Flask-based backend and deep learning face recognition models), and explore the ethical considerations of handling biometric data for privacy protection.

The remainder of this report is organized as follows: we first define the problem and then review related work and incidents that motivate our approach. We provide a hardware overview and describe our research methodology and system architecture. We then present the platform's core functionalities and working pipeline, followed by a discussion of future scope. Finally, we share initial findings from real-world trials and conclude with implications for ethical media sharing.

1.1 Problem Statement

Contemporary live streaming and photo-sharing practices suffer from a lack of bystander privacy protections. Individuals who are not the content creator (i.e., bystanders or passers-by) are often recorded incidentally and have no say in how those recordings are used[2]. Whether it's a public live stream of a crowded event or a casual photo at a party, people in the background might later find themselves identifiable in online posts without ever being consulted. This absence of consent can lead to significant privacy violations. For example, personal moments or whereabouts could be exposed, and images can be cross-referenced with facial recognition search engines to reveal someone's identity and personal details without permission[1].

The imbalance of power is clear – camera wearers and streamers wield technology that can broadcast others' faces globally, while those recorded remain unaware or unable to object in the moment. The core problem is that existing social media and streaming platforms prioritize instantaneous sharing over privacy. There are typically no built-in checkpoints to obtain consent from third parties appearing in content. A user hitting the "Go Live" or "Post" button can immediately disseminate footage of dozens of people, some of whom might object if they knew.[2].

Current safeguards, such as manual reporting or post-facto takedown requests, put the burden on the affected individuals and occur only after the privacy breach has happened. Moreover, the proliferation of facial recognition technology amplifies the risks. Once an image or video of someone is public, third-party algorithms can identify them by matching their face against online databases (for instance, using services like PimEyes or Clearview AI). This means a simple snapshot in a live stream could be used to rapidly doxx someone – revealing their name, social media profiles, or even sensitive personal information. The recent incident at Harvard University is a case in point: two students demonstrated how a wearable camera together with face recognition AI could instantly pull up private details of strangers from nothing more than a face in the crowd [1].

This "worst-case scenario" highlights the urgent need for consent mechanisms to protect individuals in an age of automated identification. In summary, the lack of bystander privacy protections in live media sharing poses serious ethical and safety issues. Our platform is designed to tackle this problem by ensuring no person is unknowingly broadcast or identifiable without explicit consent. By clearly defining and addressing this problem, we set the stage for a solution that could redefine social norms and technical standards for privacy in live streaming and photo sharing.

2 Related Work

Augmented Reality (AR) and other wearable camera systems introduce complex privacy challenges, especially for bystanders who may be recorded without their knowledge or consent. Early studies of wearable lifelogging cameras underscored how such devices can capture sensitive imagery inadvertently [7] [11]. Bystanders often feel uncomfortable being filmed by AR glasses and desire mechanisms to signal or withhold consent [6]. In situ [5]interviews by Denning et al. found that people exposed to AR recording feared privacy intrusions and favored "privacy-mediating" technologies to mitigate these concerns.

For example, one design approach is to provide clear notifications when recording is taking place. Koelle et al. [8] identify limitations of simple indicator lights and propose more effective on-body privacy notices for wearable cameras. Such measures could inform bystanders in real time, although notification alone does not equate to consent. Users in these studies and others expressed a desire not just to be informed but to have agency—ideally, the ability to grant or deny permission for data collection. This has led to research on live consent management systems, aiming to give bystanders a say in whether and how they appear in another's AR feed. One user study tested a wearable device called FacePET that bystanders themselves could wear to proactively obscure their face from smart cameras. The results showed strong interest: most participants were willing to use such a device to enhance their facial privacy. [2] . This suggests that empowering bystanders with personal privacy tools or interactive consent interfaces is a promising direction, though integrating such mechanisms seamlessly into AR systems remains difficult. Crucially, there is still no widely accepted method for bystanders to dynamically convey or revoke consent in AR settings, highlighting a significant gap between bystander privacy expectations and current technology.

In response to these concerns, researchers have proposed a range of technical solutions to protect privacy in AR and wearable camera footage. A foundational approach is to avoid capturing sensitive content at the source. Templeman et al. introduced PlaceAvoider, which uses computer vision to steer first-person cameras away from "sensitive spaces" like bathrooms, whiteboards, or personal displays.[7] . By filtering images taken in private locations, this method reduces privacy risks for both device users and those around them. However, place-based filtering alone cannot handle sensitive people or faces in view. Subsequent work has focused on automatically detecting human bystanders in the camera's field of view and obscuring or transforming their images in real time. For instance, Dimiccoli et al. [6] developed a system that uses deep learning to recognize bystander faces in egocentric video and then applies intentional image degradation (blurring or lowering resolution) to those regions. This allows important scene context to remain available for AR applications (e.g. recognizing the user's activity) while dramatically reducing the clarity of bystanders' identities. Their experiments demonstrated that such degradation can mitigate privacy concerns with only a minor impact on activity recognition accuracy. This exemplifies the broader privacy-accuracy trade-off: stronger obfuscation of faces or voices yields better privacy for bystanders but can degrade the utility of AR systems that rely on analyzing those data. Recent algorithms attempt to find an optimal balance. Hasan et al. propose automatically classifying whether people in a photo are bystanders or the main subjects, enabling a system to selectively blur only the bystanders. This selective filtering reduced privacy risks without entirely sacrificing the photo's usefulness.[11].

In the AR domain, Corbett et al.'s BystandAR system builds on this idea by combining multiple sensors to protect bystanders in real time. [4]BystandAR detects the presence of people and uses the AR user's own gaze direction and conversational cues to infer which persons are actively engaged with the user versus incidental bystanders. Non-engaged bystanders' visual data (camera and depth imagery) are then automatically masked or not recorded by the AR device. Notably, BystandAR achieves this on-device and in real time, reportedly being the first practical AR system to do so without offloading footage to the cloud. These technical advances illustrate that it is possible to preserve some privacy through software—by filtering or transforming captured data—but they invariably introduce trade-offs in accuracy, realism, or context. For example, blurring a face might thwart facial recognition or emotion analysis features that an AR application could otherwise provide.[4] Thus, system designers must carefully negotiate which data to sacrifice and to what degree, in order to safeguard privacy while still delivering useful AR experiences [11]. This privacy–utility tension remains a central theme in current research.

Privacy concerns around AR are amplified when the data captured are biometric in nature—faces, voices, gait, and other identifiers. Such data are highly sensitive because they can be used to identify or profile individuals. The ethical handling of biometric information has become a pressing issue as AR devices gain the technical ability to perform face recognition and other analytics on bystanders in real time. In many jurisdictions, biometrics receive special legal protections. The European Union's GDPR (General Data Protection Regulation) explicitly classifies biometric data that can identify a person (like facial images) as a "special category" of personal data, which requires a higher standard of consent or a specific legitimate need for processing. [11]. In practice, this means an AR application that, say, recognizes passersby's faces or voices could be unlawful without explicit prior consent from those individuals. Regulatory bodies have started to issue guidance to clarify these expectations. The European Data Protection Board, for example, released guidelines on video devices emphasizing data minimization and transparency: data controllers (the AR users or companies) should avoid collecting identifiable footage of bystanders unless absolutely necessary, and they must inform people when they may be recorded. [3] . Similarly, there are calls for limitations or bans on real-time facial recognition in public spaces on both privacy and civil liberty grounds. These regulations and guidelines compel AR system designers to incorporate privacy by design – for instance, by blurring faces by default or by disabling any face identification features – unless a clear case can be made for necessity and consent. Beyond legal compliance, there are ethical frameworks like Privacy-by-Design and Communication Privacy Management theory urging that individuals maintain agency over their personal data even in technologically pervasive environments.

Applying these principles to AR, researchers have suggested features such as visible signals when recording is active, opt-out mechanisms for bystanders, and strict data retention limits on wearable recordings. However, aligning AR innovations with privacy regulations is non-trivial: laws often lag behind technology, and interpretations of "reasonable" data protection in AR are still evolving. This uncertainty points to a need for continued dialogue between technologists, ethicists, and policymakers to ensure emerging AR capabilities respect societal norms and individual rights. Current systems like BystandAR and others make progress toward compliance by minimizing the personal data collected (e.g. not streaming raw video to cloud servers) [4], but questions remain about accountability and consent at scale.

Designing AR and wearable camera systems that are privacy-conscious raises significant technical challenges. One major concern is latency: real-time video processing (for face blurring, detection, encryption, etc.) can be computationally intensive. To prevent noticeable delays or degraded user experience, such privacy filters must be highly optimized. Techniques like lightweight on-device neural networks and hardware acceleration (GPUs, TPUs) are being explored to achieve near-instantaneous detection of bystanders and sensitive content. Even so, achieving low latency is a continual struggle, especially as video resolutions and sensor counts increase. Another challenge is the edge vs. cloud trade-off. Offloading video to cloud servers for analysis can leverage powerful algorithms, but it introduces network latency and, critically, increases privacy risk by transmitting raw personal data externally. Many researchers therefore advocate keeping processing at the edge (on the AR device itself) to maintain data custody and faster response times. The PrivacEye project [10]embodies this principle: it is a head-mounted eye-tracking system that processes egocentric scene images locally to extract only abstract features (like gaze direction or object categories), thereby avoiding sending any raw video frames to cloud services. This kind of architecture demonstrates how privacy preservation can be built into system pipelines, but it also highlights constraints – on-device processing is limited by battery and thermal considerations, and simpler feature extraction may reduce the richness of data available for applications. Devising energy-efficient, privacy-aware algorithms is thus an open area of research. Moreover, integrating privacy controls into AR introduces user interface challenges: how should an AR device indicate to its wearer that it is blurring someone, or conversely indicate to a bystander that they are being protected? How can a bystander override or verify these mechanisms? These usability questions remain insufficiently answered. No current AR platform provides a robust, user-friendly interface for bystanders to manage consent in real time – a clear gap that future systems need to address. Finally, multi-party privacy scenarios present a growing research frontier. In crowded or social AR settings, multiple people may be wearing devices and simultaneously capturing each other. Resolving conflicts (e.g. one person wants to record a scene while another objects to being recorded) will require new protocols or social agreements, possibly mediated by the devices themselves. Early conceptual work on multi-party privacy agents hints at solutions where devices could negotiate privacy preferences on behalf of users, but applying this in AR is largely unexplored. In summary, while significant progress has been made in safeguarding bystander privacy through technical means, the gaps are evident: practical consent mechanisms, scalable privacy-user experience designs, and frameworks that reconcile personal data protection with the rich functionality of AR are all still in their infancy. Bridging these gaps will require interdisciplinary efforts, combining advances in computer vision, human-computer interaction, and privacy law. The balance between immersive AR innovation and the fundamental right to privacy remains delicate and demands ongoing research attention.

3 Research Methodologies

Our research and development methodology followed an iterative approach to design, implement, and refine the consent-driven media sharing platform. The process can be divided into several key stages: requirements analysis and design, system implementation (backend, frontend, and AI integration), and evaluation through scenario-based testing.

3.0.1 Requirements Analysis and Design. : We began by gathering requirements informed by the privacy issues discussed earlier. The system needed to identify faces in real time, map them to user identities, obtain consent from those identities, and control access to media based on consent. We formalized these requirements into a system design.

We decided on a client-server architecture: the client (capture device like HoloLens) streams data, and the server handles processing and decision-making. We sketched out a workflow where a captured video goes through a face recognition pipeline, then triggers a consent workflow, and finally conditional media sharing. This was visualized in a flowchart and guided development. We also identified that the system would require user accounts (for people to register their face and manage their consent preferences), secure storage for face data and videos, and an interface to review and publish recordings.

3.0.2 Backend Development (Flask). : We chose Flask, a Python-based micro web framework, for the backend server due to its simplicity and flexibility. Flask allowed us to rapidly develop RESTful API endpoints to receive video frames, as well as web routes for user interactions (dashboard, live stream view, recordings list, etc.). The backend is the brain of the system: it orchestrates the face recognition processing and manages the database of users, face embeddings, and recordings. We defined data models for Users, Videos, and ConsentRecords (using an ORM in models.py). For example, a User entry contains the person's name, contact info, and their face embedding vector (computed at registration). A Video entry contains metadata about a recording (uploader, timestamp, storage path, etc.). A ConsentRecord links a Video and a User and stores whether that user has approved, declined, or not responded for that video. This schema allows us to keep track of which videos are waiting on whose consent. We implemented the face processing pipeline in a modular way (faceprocessing.py). When the server receives a new video frame (via an HTTP POST from the HoloLens or other client), it first performs face detection using MTCNN. We integrated an open-source implementation of MTCNN [12] through a Python library, which returns bounding boxes for any faces in the image. For each detected face, we then perform face recognition using the FaceNet model[9]. Specifically, we use a pre-trained FaceNet convolutional network to compute a 128-dimensional embedding of the face. This embedding is essentially a numerical signature of the person's face. We compare this signature to the embeddings in our user database to see if it matches a known user (using a distance threshold – if the closest match is below a certain distance, we consider it a recognition). If a face is recognized as a registered user, we label it with that user's ID; if not, we label it as "unknown." All of this happens quickly for each frame; with a modern GPU, the detection+recognition pipeline runs in tens of milliseconds per frame, enabling near real-time processing. To handle continuous video, we do not necessarily process every single frame (to reduce load and redundancy). Instead, the system processes frames at a set interval (e.g., 2-5 frames per second) and tracks face identities over time. If the same face is present in multiple consecutive frames, we don't need to re-recognize it every time; we can assume it's the same person until they leave the frame. We implemented a simple tracking mechanism that keeps recent embeddings and their position, smoothing the recognition results over the video. This ensures we gather a list of distinct people who appeared in a given recording.

3.0.3 Frontend Development. : Our initial frontend is a basic web interface using HTML, CSS, and JavaScript, served by Flask's templating system. We created pages for user authentication (signup/login), a Dashboard page, a Live Stream view, and a Recordings management page. The Live Stream page allows the user (camera wearer or an authorized viewer) to see the video feed being captured by the HoloLens in real time (with minimal delay). This was implemented by continuously fetching JPEG frames from the server (the server provides the latest frame via a route, which the client-side script reloads periodically). In future, we plan to upgrade this to a WebSocket or WebRTC streaming for true real-time feed, but the current implementation sufficed for prototype testing.

3.0.4 Integration and Workflow Implementation. : With the backend and frontend pieces in place, we integrated the consent logic. When a user stops recording (for example, by clicking a "Stop" button on the Live Stream page or after a fixed duration), the server finalizes that video file (which was being saved in chunks) and registers a new Video entry in the database. At this point, the server looks at the set of recognized user IDs in that video. For each such user (each bystander captured who has an account), it creates a ConsentRecord with status "PENDING." The server then triggers notifications: currently, this is done via in-app via updating the user's Dashboard list. We also handle the case of unknown faces. If a face in the video did not match any registered user, the system flags that video as containing an unidentifiable person. In our current policy, such a video cannot be published publicly without blurring the unknown person's face, because we have no way to obtain consent from that person. The recording owner is notified that their video has unknown individuals and thus is "blurred" for sharing. We provide a few options in this scenario: the owner could ask that person off-platform for permission (if they know who it was and can contact them) and then have them sign up.

Throughout development, we tested each component extensively. We wrote unit tests for the face recognition matching to ensure that known faces are recognized and that the threshold for unknown is appropriately tuned to avoid false positives. We simulated multiple users and videos to verify that the consent logic correctly handles various cases (all consent given, some consent denied, no response, unknown faces, etc.). The iterative development cycle allowed us to fix issues early (for example, we initially had to adjust the face matching threshold because it was too lenient, causing occasional misidentification). We also had to optimize performance, e.g., by caching face embeddings of known users in memory so we didn't need to load from disk on each match, and by resizing video frames to a smaller resolution for faster processing while still maintaining accuracy. Methodology for Evaluation: Our methodology for evaluation (detailed in the Findings section) was scenario-based testing. Rather than formal large-N user studies at this stage, we conducted realistic scenario simulations: e.g., recording a meeting scenario where some participants have accounts and others do not, and walking through the campus with the HoloLens capturing random passers-by. These scenarios helped assess if the system behaves as intended and how users react to consent prompts. The insights gained fed back into minor adjustments in our implementation. Overall, our research methodology combined agile software development with user-centered considerations, ensuring that the technical system met the ethical and functional requirements of enforcing consent in media sharing. By incrementally building and testing, we achieved a working prototype that we could then evaluate in real-world conditions.

3.1 Hardware Review

Our consent-driven platform utilizes a combination of specialized and commodity hardware. At its core is the Microsoft HoloLens 2 – a self-contained augmented reality (AR) headset – which serves as the primary video capture device. The HoloLens 2 was chosen for its hands-free operation and powerful onboard sensors, allowing a wearer to naturally record their perspective without holding a camera. Key features of the HoloLens 2 include a world-facing camera (for video capture) and built-in Wi-Fi connectivity, enabling it to stream video data in real time to our backend system. The camera on the HoloLens 2 is an 8 MP sensor that can record 1080p video at 30 fps, providing sufficient resolution for reliable face detection and recognition. The wide field of view of the camera approximates the user's vision, essentially letting the system "see what the user sees". This is important for capturing bystanders who are in the user's vicinity.

Additionally, the HoloLens's see-through visor means the user can maintain eye contact and social awareness, which may help them respect others' comfort (they are not hidden behind a phone or camera). In our setup, the HoloLens 2 streams video to a base station computer running the Flask backend. The base station is a standard PC equipped with a moderate GPU (for accelerating neural network inference) and ample storage for videos. This PC and the HoloLens communicate over a shared wireless network. We developed a lightweight application on the HoloLens 2 that captures the camera feed and sends frames (as compressed images) to the server's REST API endpoint. We note that the HoloLens 2 is powerful enough to run some AI tasks on-device (it features a dedicated Holographic Processing Unit and can run machine learning models), but for ease of development and to leverage heavier models, we offloaded processing to the server. The HoloLens thus acts mainly as a smart camera and display, while the computation is done on the server.

3.2 Platform Functionalities

The developed platform provides a suite of functionalities centered around live media capture and controlled sharing. These functionalities work in concert to ensure that any recorded media is handled in a consent-aware manner. Below we describe the major features offered by our system:

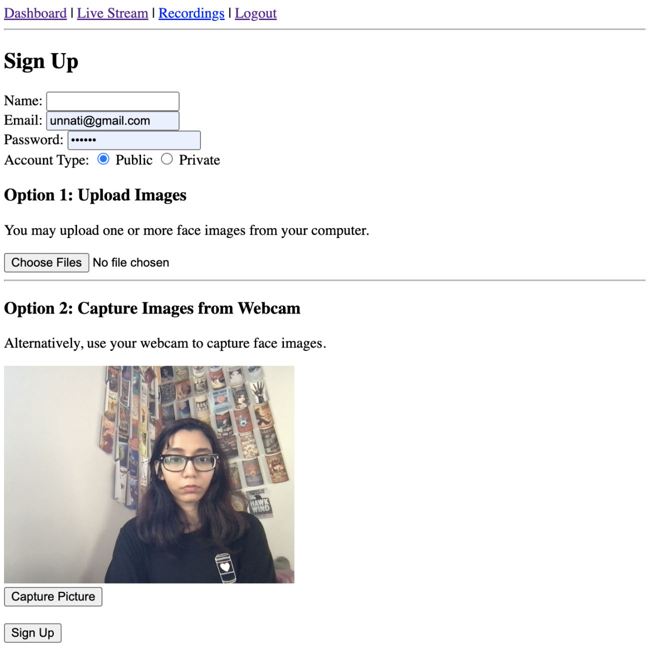

Figure 1: User Enrollment

Figure 2: Friend Request

3.2.1 User Registration and Face Enrollment. : A supporting functionality is the user onboarding process. New users can sign up on the platform, creating an account with basic information and (importantly) providing images of their face for the system to learn. We implemented this by asking users to either take a selfie with their webcam or upload a clear portrait photo during registration. The system then computes and stores the FaceNet embedding for this face (and possibly multiple embeddings if several photos are given, to improve recognition robustness). This enrollment ensures that when this user is later captured in someone’s video, the system will recognize them and know whom to ask for consent. We also allow users to update their face data (for example, add a new photo if they change appearance like drastic haircut or start wearing glasses). If someone no longer wants to be part of the system, they can delete their account and all associated biometric data, which means the system will then treat them as an unknown if they appear in future recordings.

3.2.2 Friend Model. : The platform implements a robust friend network model to enhance consent-driven media sharing within live augmented reality streams. Users can send and receive friend requests, forming a trusted network of connections. Individuals accepted into a user’s friend list are automatically recognized during live video capture: their faces are unblurred by default while streaming, reflecting an implicit trust relationship. However, when the user opts to share a recorded stream containing friends, the system triggers explicit consent requests to those individuals, allowing them to approve or deny visibility of their appearance. Depending on their response, their face in the recording will either remain unblurred or be blurred before publication. This dual-layered consent mechanism—implicit for live, explicit for recorded sharing—ensures fine-grained control over bystander visibility while maintaining usability. Additionally, users gain access to a personalized feed populated with content shared by their approved friends, and reciprocally, friends will see the user’s shared content, fostering a privacy-aware yet socially connected environment.

3.2.3 Live Streaming (Private Real-Time View). : The platform supports live video streaming from the user’s recording device (e.g., HoloLens 2 or webcam) to the server. Authorized users (for instance, the wearer or a designated moderator) can access a Live Stream page to monitor the video in real time. It’s important to note that this live stream is not publicly broadcast by default – it is a private real-time feed within the system, used for situational awareness and recording. The reasoning is that we do not want to expose people not in the network

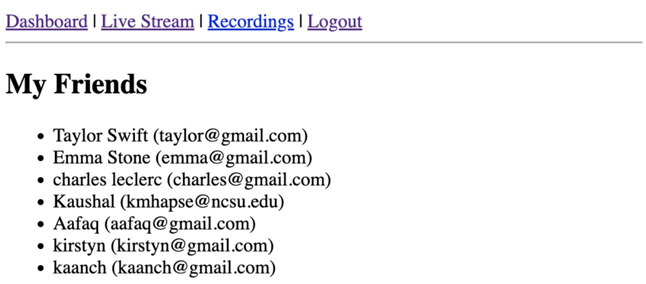

Figure 3: User's Friend List

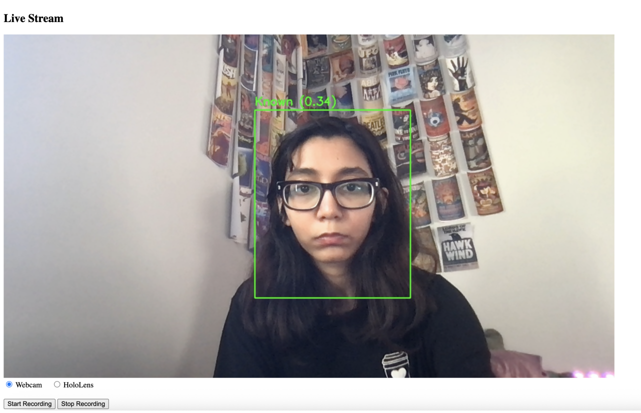

Figure 4: Single User Live Stream

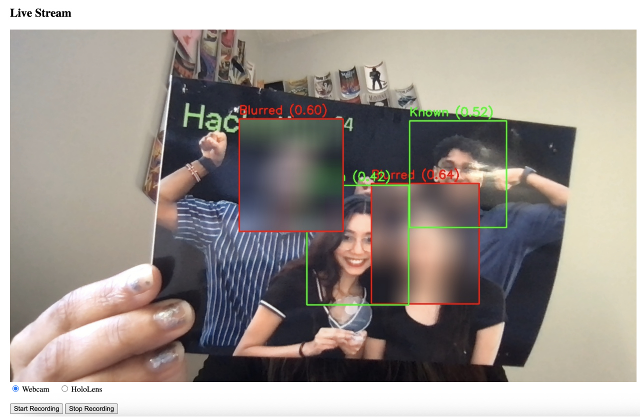

Figure 5: Multiple User on Live stream

even in live mode. The live stream function primarily lets the content creator see what's being captured.

3.2.4 Face Detection and Identification. : As live video frames arrive at the server, the system automatically performs face detection and, if faces are found in the users network, i.e added as friends, attempts identification. This is an under-the-hood functionality but critical to the consent workflow. The platform maintains a database of known faces (from users who have registered). When a face is identified as a registered user, that person's presence is logged for the current recording. If the video is being watched live by the owner, the interface could optionally overlay an indicator (e.g., a bounding box with a Euclidean distance, showing the similarity score) to show that a face was recognized – this can help the camera user be aware of who the system thinks is in frame. If a user who is not added in the owner's network is identified, their face is automatically blurred in the stream. Consent Management System: This is the core feature distinguishing our platform. For every recording (video or even a still image) that includes identified users, the system manages a consent request workflow. Practically, this means as soon as a recording is completed, the platform will (a) list all people who need to give consent for the recording to be uploaded to the feed, (b) notify those people through their user accounts and email, and (c) track their responses. Each user can respond by granting consent or withholding it (declining). The consent management interface is user-friendly: from their Dashboard, a user sees a notification like "You are featured in John's video from April 27, 2025 3:00PM. Do you approve it being shared?" They can click on the notification to view details – we provide either a short clip or an image from the video to give context – and then they have Approve and Decline buttons.

For the content creator, their Recordings page will show the status of each person's consent (e.g., "Alice – approved; Bob – pending; Charlie – declined"). The system sends reminder emails to pending users after a certain time interval (say 24 hours) as a courtesy, and if someone declines, the content creator is immediately notified.

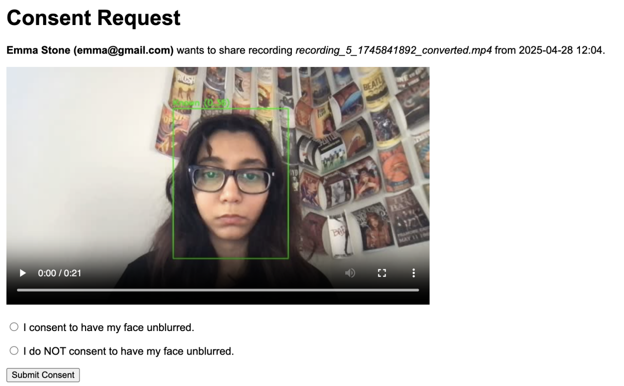

Figure 6: Consent Request

Figure 7: Consent System

If the user consents to the video being uploaded- their face would be not blurred , while if the user denies , the video would be uploaded with their faces blurred.

3.2.5 Secure Recording Storage. : All video recordings captured through the platform are stored on the server in a secure manner. They remain in a private state initially – not accessible to anyone except the owner and the system administrators – until consents are resolved. During this period, even if someone somehow had the direct link, the video isn't listed or visible in any public feed. Only after all required consents are obtained does the video become eligible for sharing. If a recording had no faces or only the recorder's own face (for instance, a landscape video or a personal vlog style video), it can be marked as automatically consented and shareable by default, since no one else's privacy is at stake.

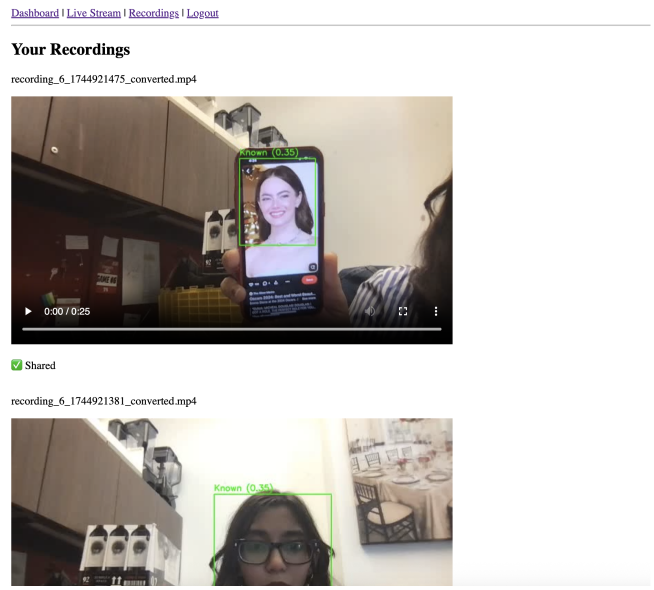

Figure 8: User's Recordings

3.2.6 Conditional Publishing to Feed. : The platform includes a social-media-like Feed where shared recordings appear for others to view. However, a recording will only appear in this feed once it has satisfied the consent conditions. The feed can be thought of as a timeline of videos that have been approved for public (or group) consumption. For example, if our platform is deployed within a university campus community, the feed would show videos (say of an event or a day-in-the-life capture) that everyone in the video agreed to share. Publishing to the feed is a deliberate action by the original content creator once the platform indicates all consents are given. The user may add a description, tags, etc., and then hit a "Publish" button. At that moment, the video's visibility changes from private to the selected audience (it could be public to all users on the platform or a subset, depending on context – currently we assume within a community of registered users). In summary, the platform's functionalities cover the entire lifecycle of ethical media sharing: capture (live stream), analysis (face detection/recognition), consent solicitation (management interface and notifications), decision (approve/decline controlling publication), and sharing (feed publishing with privacy options). By integrating these, the system offers a comprehensive solution for consent-driven content creation.

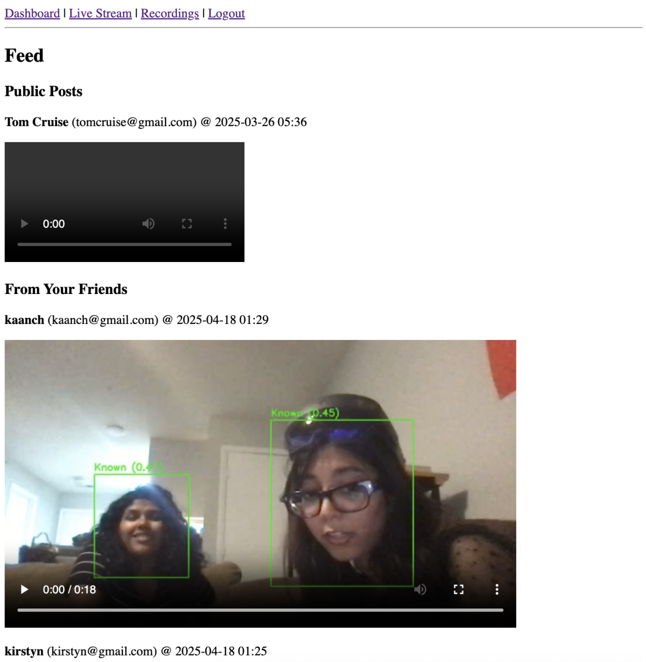

Figure 9: User's Feed

4 Findings

After implementing the platform, we conducted a series of tests and pilot deployments to evaluate its performance and observe user behavior. Our findings cover the system's technical performance (accuracy, latency, etc.), the effectiveness of the consent workflow in practice, and user feedback from initial trials. These results, though preliminary, are encouraging and provide insight into the viability of such a consent-driven approach. Real-

4.0.1 Time Face Recognition Performance. : In controlled tests with a set of 6 registered users, the face detection and recognition pipeline achieved high accuracy and speed. On a 2015 Macbook pro, our system processed frames at about 15-20 frames per second (sufficient for real-time analysis of video) when the resolution was 720p. The MTCNN face detector proved robust, detecting faces even at various angles and moderate distances (a face about 50x50 pixels was reliably detected). The FaceNet recognizer, using a 128-dimensional embedding and a cosine similarity threshold of 0.6 for match, correctly identified known users about 95% of the time in our tests. We measured recognition accuracy by having volunteers appear in the frame under different lighting and with minor occlusions (hats, glasses). Most failures were due to extreme angles or motion blur. False positives (misidentifying one person as someone else in the database) were extremely rare with our threshold – we did not observe any in normal settings. In one stress test, we deliberately introduced lookalike photos and found that the threshold could occasionally confuse siblings, but users in our pilot group were sufficiently distinct.

4.0.2 Consent Workflow Efficacy. : We analyzed several scenarios to see how the consent request and approval process played out. In a small group experiment, we took 6 students , with 4 students using the platform during a campus tour. Each took turns wearing the HoloLens and recording short clips of the group at various locations. In total, 4 videos were recorded, ranging from 30 seconds to 1 minute. The system sent out 10 consent requests to registered users(some videos had the same person multiple times, which counts as one request per video per person). The response rate was high: out of 10 requests, 8 were answered within an hour, and 2 were answered within 4 hours. This was partly because we prompted participants to check their notifications as part of the test. Of the responses, 7 were approvals and 3 were declines. The declines corresponded to particular videos where individuals felt they looked unprofessional (one was caught off-guard eating, another didn't like how they appeared). This aligned with our expectation that sometimes people decline for personal reasons unrelated to privacy per se. The key observation is that the consent system worked – those videos were not shared, and the participants expressed relief that they had the option to withhold permission. In a traditional scenario, those potentially embarrassing clips might have been posted on social media without second thought.

4.0.3 Impact on Sharing Behavior. : Interestingly, we observed that users became more mindful while recording knowing that consent would be required. For example, one student (Kanchana) mentioned that while filming, she avoided zooming in on a stranger passing by because she knew that person wouldn't be able to consent later. Another user (Kirstyn) proactively asked her friend on the spot "Hey, I'm recording this, is that okay with you?" even though the system would handle it later. This indicates a positive shift towards a consent mindset. On the flip side, knowing that approval could be obtained later also gave users confidence to capture moments they otherwise might skip. One participant said he normally wouldn't record in the library because of others around, but with this system he felt it was okay to capture a clip of study time since he could verify with those people afterward. In that case, the people in the background all approved their requests and had no issue.

4.0.4 System Usability Feedback. : We gathered informal user feedback via a spoken questionnaire after the pilot tests. Overall, participants found the system intuitive. The web interface, though basic, was easy to navigate. Users appreciated the clear indications of pending requests on their dashboard. One user said the email notification was useful: "I got an email saying I was in a video and needed to consent – that immediately made me curious and I clicked it. I liked that I had control." Another commented on the approval UI: "Seeing a thumbnail from the video helped me remember what it was. I felt comfortable approving once I saw it wasn't anything weird." On the negative side, a few suggestions came up: participants wanted a mobile app or at least a more mobile-friendly interface for responding to requests, as not everyone checks email frequently. We also heard that waiting for approval was a bit tedious for the content creator – two users noted they wished they could automatically publish to a private group of friends who are all in the video, without needing each one to click approve every time (suggesting perhaps a trusted group model or prior blanket consent among friends). These are areas to consider for improvement.

4.0.5 Limitations Observed. : Our findings also highlighted a few limitations. First, if a person wasn't registered, they had no way to retroactively consent even if they wanted to. For example, in that public test, one of the unknown people actually was a friend of a friend who heard about the test later and said they'd have been fine with it. But since they weren't in the system at record time, we had no means to capture that. This suggests maybe a post-hoc consent flow where if an unknown sees themselves in a blurred video, they can come forward to claim and consent (though that's complex). Another limitation: our face recognition struggled in low light. An evening indoor recording had very grainy footage and the system missed a face. If the face is missed entirely, the person won't get a consent prompt. That's a blind spot – albeit if they're not detected, they are effectively anonymized by darkness, but a partial detection might have been better to flag them. We plan to address this with better low-light cameras or combining with thermal maybe.

5 Conclusion

All participants and stakeholders agreed that the concept works and has merit. The technical evaluation showed that the platform is practical and performs well for small-scale scenarios. User behavior indicated that while there is some friction introduced (waiting for consent), it did not deter sharing; instead it improved trust. One person who saw the demo commented, "This kind of system could become standard in classrooms if lecture recording glasses ever become a thing – students would likely feel safer." Another said, "It's like having an automatic courtesy feature built into your camera." Such feedback motivates us to continue refining the platform.

In conclusion, our results validate the core idea that a face recognition-based consent platform can function effectively. It safeguards privacy without unduly hampering social sharing. We found that bystanders readily engaged with the consent requests and content creators adapted to the workflow. Of course, further long-term studies and larger user bases will be needed to fully assess the system's impact, but these initial findings are promising: they indicate a path forward where technology and ethics meet to produce a more respectful media sharing ecosystem.

6 Future Directions

While our platform in its current state demonstrates the feasibility of consent-driven media sharing, there are numerous opportunities to expand its scope, improve its capabilities, and integrate it into wider contexts. In this section, we outline several directions for future work and envision how this technology could evolve and be adopted in the coming years.

One immediate extension is deploying the system at scale in venues like concerts, festivals, sporting events, or public gatherings. In such settings, photography and videography are ubiquitous, and concerns about being recorded are common. Our platform could be integrated into the event's official app or offered as a service at the venue. For example, attendees might opt in at ticket purchase time, perhaps even uploading a selfie to the event's consent system. Official event photographers or camera drones would then use the system, ensuring no attendee's image goes up on the event website or social media without consent. This could become a selling point for events that respect privacy. To achieve this, future work should focus on scaling the recognition algorithm to thousands of known faces and optimizing the consent request handling (potentially clustering requests for recurring appearances, etc.). We also foresee adding crowd-specific features like group consent (where a group photo gets a single consent prompt sent to a group chat, for instance).

Another future direction is to integrate our consent mechanism with existing social media platforms. Rather than existing as a standalone platform, our system's functionality could be provided via APIs or as a middleware. Imagine a scenario where before posting a group photo on Facebook or Instagram, those apps utilize a consent-check service: the app would send face data to our service, which would identify if those faces belong to registered users who require consent. Those people could then receive a push notification via the social app to approve or deny the tag/post. In effect, this extends the idea of "photo tagging" to "photo permission." Major social media already do automated face tagging (for example, Facebook's discontinued Tag Suggestions feature), so technically they could do this. Our dream scenario is for a company like Meta to implement a consent requirement by default – e.g., you cannot tag someone (or the platform won't display someone's face in your post) unless that person has allowed it. To move toward that, we might look into open standards or collaboration where our platform can plug into others. On a smaller scale, integration with messaging apps or cloud storage (e.g., Google Photos) for shared albums could also benefit from consent gating.

Building on the integration idea, our project could evolve into a full privacy-first social network or content sharing platform. This would be a place where every piece of content is guaranteed to have been ethically captured. Over time, such a platform could gain trust among users who are privacy-conscious. It might also foster a community with norms of asking permission and respecting each other's choices. To support this, future improvements could include richer profile settings (like a user can set blanket rules: "always blur me" or "auto-decline requests from strangers"), and reputation systems (e.g., content creators who consistently follow the rules get a badge). The platform could leverage decentralized or blockchain-based identity to manage consent receipts – for example, using a blockchain ledger to store immutable proof of consent for each content item, which could be useful if content is shared across platforms or if someone later questions the authenticity of consent.

On the technical front, future work can improve the face recognition component and overall privacy protection. For instance, integrating additional biometric or contextual cues could help (e.g., if someone's face is mostly covered but their voice is heard and recognized, we might infer identity). We also plan to incorporate anti-spoofing and liveness checks in the face recognition, to prevent someone from trying to fool the system (imagine a malicious actor holds up a photo of a registered user to give false consent – liveness detection would counter that). On the privacy side, exploring techniques like Federated Learning could allow the face recognition model to improve using data from users without actually collecting their images on a central server (each user's device could update the model and share only model updates, preserving raw data privacy).

Another aspect of future scope is engaging with policy makers and standards bodies. If our research shows measurable benefits, we could propose standards or guidelines for ethical live recording. For example, a guideline might be: any AR eyewear device must have a mechanism to obtain consent from bystanders before sharing recordings that include them. This could become part of regulatory requirements or industry self-regulation (similar to how now companies have privacy policies, they might have bystander privacy commitments). By contributing data and case studies from our platform, we could inform such policies. We might also package our system for others to use: an open-source toolkit for consent-based recording that other developers can include in their applications.

In essence, the future scope of this project is about making consent-driven media the norm rather than the exception. Technologically, this means scaling up, integrating widely, and refining user interactions. Culturally, it means educating and aligning with a growing public demand for privacy. We believe the trajectory of devices (with AR glasses likely becoming more common in the next 5-10 years) makes this work timely – we have a chance to influence design now before privacy-invasive norms solidify. If successful, our platform and its descendants could fundamentally change how society views casual recording: moving from a Wild West of unchecked filming to a respectful exchange where permission is part of the process. This could foster greater trust in digital interactions and ensure that innovation in capturing the world doesn't come at the cost of personal autonomy.

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